Chapter 16 exercises

These 22 problems cover every source subsection, 16.1 through 16.13. Specify the group, code, or action used in each argument.

16.1 Groups

1. Decide each statement with conditions and witnesses. Is $GL_1(\mathbb{R})$ abelian? Is $GL_2(\mathbb{R})$ abelian? If not, exhibit two invertible noncommuting matrices. For $n \geq 2$, is $(\mathbb{Z}_n, \cdot)$ a group? For prime p, which residue subset forms a multiplicative group?
2. Prove uniqueness of group identity and inverse, then prove left cancellation. Find the order of 3 in $(\mathbb{Z}_{12}, +)$.

16.2 Homomorphisms and cyclic groups

3. For $\varphi : \mathbb{Z}_{18} \to \mathbb{Z}_6$, $[k]_{18} \mapsto [k]_6$, prove representative independence and the homomorphism property. Find kernel, image, and the quotient's structure.

16.3 Cosets and normality

4. In $(\mathbb{Z}_{12}, +)$, take $H = \{0, 4, 8\}$. List distinct left cosets, prove they partition the group, and compute the index using Lagrange.
5. Let $G = S_3$, $H = \langle (12) \rangle$, $g = (13)$. List gH, Hg, determine normality, and use two representatives of the same coset to show how proposed quotient multiplication can depend on representatives. Compose permutations right to left.
6. Prove both directions: for $H \leq G$, multiplication $(aH)(bH) = (ab)H$ is well defined for all representatives if and only if $H \trianglelefteq G$.

16.4 RSA

7. Take $p = 5, q = 11, e = 17$. Find $n, \varphi(n), d$, and encrypt and decrypt $M = 7$, displaying intermediate modular squares.
8. Use the same keys with $M = 15$, which is not coprime to n. Compute encryption and decryption completely with CRT. Do not apply Euler's theorem without its coprimality condition.

Chapter 16 exercises (continued)

16.5 Detection and correction

9. For $C = \{000, 111\}$, state detection and correction capabilities on a binary symmetric channel. If 111 is sent and 100 received, analyze nearest decoding and why correcting one error does not simultaneously guarantee detection of every two-error event under that decoder.

16.6 Hamming metric

10. Find all three pairwise distances for $x = 01010, y = 11001, z = 00011$ and check the triangle inequality. Compute $V(5, 2)$. How many errors does distance 5 guarantee can be corrected?

16.7 Generator and parity-check matrices

11. Work over $\mathbb{F}_2$ with

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}, \qquad G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

Verify $GH^T = 0$, encode message 1010, and prove minimum distance 3 using the columns of H.

16.8 Coset leaders

12. For $C = \{000, 011, 101, 110\}$, $H = [1\ 1\ 1]$, list its two cosets, syndromes, and every possible coset leader. Does it uniquely correct every single-bit error?

16.9 Hamming matrices

13. Given codeword $c = 1010101$ of Problem 11 and received word $r = 1110101$, use the syndrome to locate and correct the error. Verify from sphere packing that the $[7, 4, 3]$ code is perfect for one-error correction.

16.10 Burnside

14. Color four square vertices with three named colors and act by all eight symmetries D_4. List fixed counts by motion type and find the orbit count. Explain the denominator through counting fixed pairs in two ways.

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Chapter 16 exercises (continued)

16.11 Cycle index

15. Give cycle indices for the natural square-vertex actions of C_4, D_4, and evaluate their three-color orbit counts.

16.12 Polya

16. For a three-position rotational necklace C_3, use colors r, w, b to derive the full pattern inventory. Explain $2rwb$, and check its total coefficient sum against Burnside.

16.13 Integrated checks

17. Prove or refute each statement: (a) $GL_1(\mathbb{R})$ is nonabelian. (b) Multiplication on all of $\mathbb{Z}_p$, including zero, is a group for prime p. (c) Every subgroup automatically defines quotient group multiplication. (d) Zero syndrome proves transmission had no error. (e) Burnside's denominator is $|X|$.

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Chapter 16 exercises (continued)

Transferring group and coding structures

18. List every additive homomorphism $\mathbb{Z}_{18} \to \mathbb{Z}_{12}$. Give $\varphi(1)$, kernel, image and surjectivity for each. The natural reduction map is not well defined for these moduli.
19. For $m, n \geq 1$, prove that $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic if and only if $\gcd(m, n) = 1$. Compare $\mathbb{Z}_6 \times \mathbb{Z}_3$ with $\mathbb{Z}_{18}$. In $\mathbb{Z}_{18}$, list every generator and every solution of $6x = 0$.
20. Extend every word of this chapter's $[7, 4, 3]$ code by the parity bit $c_8 = \sum_{i=1}^{7} c_i \pmod 2$. Prove the new dimension and minimum distance. Let s be the original syndrome on the first seven positions and p the total parity of all eight. Assuming at most two errors, give the decoder decisions for all four combinations of zero or nonzero s and zero or one p. Which decisions lose their guarantee if this assumption is removed?
21. Color six labeled circular positions red or blue, with exactly three positions of each color. Count orbits under C_6 rotations and under D_6 rotations and reflections by extracting coefficients from cycle types. Explain why directly dividing $\binom{6}{3}$ by six fails.
22. Let $C_4 = \langle r \rangle$ act on the two diagonal vertex sets $P = \{\{1, 3\}, \{2, 4\}\}$. Color each whole set red or blue. List all four fixed counts, count orbits and find the stabilizer of a configuration with one red set and one blue set. Does Burnside still apply when an action is not faithful, so distinct group elements can induce the same permutation? Explain.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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